YS-K01
Door controller

MS User Manual

Suzhou Eshine Elevator Components Co., Ltd.
Address: No.269 Yuexi road, Lili Town,
Wujiang District, Suzhou City, Jiangsu Province, China
Telephone: +86-512-82060701
Http:// www.eshine.com.cn
E-mail： Oversea.sales@eshine.com.cn

Notice:

1. In factory, the controller has been set the parameters. In generally the user don`t need to change parameters.

2. Default control mode is terminal mode (P0.06=1), if you need to use manual mode (press OD/CD to open/close door), please set P0.06=0.

3. Default door arrived contact is normally closed (P6.00=1), if you need normally opened, set P6.00=0.

4. The situation that you can`t change the parameters:

   A: Make sure you input the right password, default with no password. Check chapter 4 or contact the factory for help if you forget the password. The lock LED on panel YS-P02 can help you to judge whether the controller has password.

   B: When the motor is running, you can`t change some parameters.

      a: In terminal mode(P0.06=1), you can pull out the OC/CD/SS/COM terminal, and reset the power.

      b: In manual mode(P0.06=0), you can press OD /CD, when the door in middle place , press STOP, then you can change parameters.

CONTENTS

Chapter 1 Installation and Wiring

Danger

- Do not install if the controller is imcomplete or impaired.
- Make sure that the controller is far from the explosive and combustible things.
- Bare metallic part of the power terminal wiring should be with insulating tape to wrap up.
- Only qualified electrical engineer can perform wiring job.
- Only when the power supply is completely cut-off 10 minutes later can you do the wiring job.
- The communication wire must use shielded twisted-pair cable with 20—30mm twisted interval, and the shielding layer must be grounded.
- The encoder must use a shielded wire, and the shielding layer must be connected to ground reliably.
- Do not connect the power input terminals to the output terminals (U/V/W) of the controller.
- The earth terminal PE of the controller must be reliable earthing. It must use two separate earth wire due to the leakage current from the controller to ground.
- Do not touch the wire terminals of the controller when it is live. The main circuit terminals is neither allowed connecting to the enclosure nor short-circuiting.
- Only when the controller terminal cover has been fitted can you switch on AC power source. Do not remove the cover after power is switched on.

Warning

- Do not play metal into the controller when installing.
- Do not start and shutdown the controller by connecting and disconnecting the contactor.
- Detection of signals during the operation shall only be conducted by qualified technician.
- Do not do dielectric strength test on the controller.

Ensure the installation site meeting the following requirements:

- Do not install at the direct sunlight, moisture, water droplet location;
- Do not install at the combustible, explosive, corrosive gas and liquid location;
- Do not install at the oily dust, fiber and metal powder location;
- Be vertical installation on fire-retardant material with a strong support;
- Make sure adequate cooling space for the controller so as to keep the ambient temperature among - 10—+ 40℃;
- Install at where the vibration is lesser than 5.9m/s^2 (0.6g).

Note:

1. We need derate value if the controller operation temperature exceeds 40℃. The derate value of the output current of controller shall be 2% for each degree centigrade. Max. allowed temperature is 50℃.

2. Keep ambient temperature among -10－+40℃. It can improve the controller operation performance if install at the location with good ventilation or cooling devices.

1.1 Terminal Layout

The connection terminal layout of controller is shown as Figure 1-1.

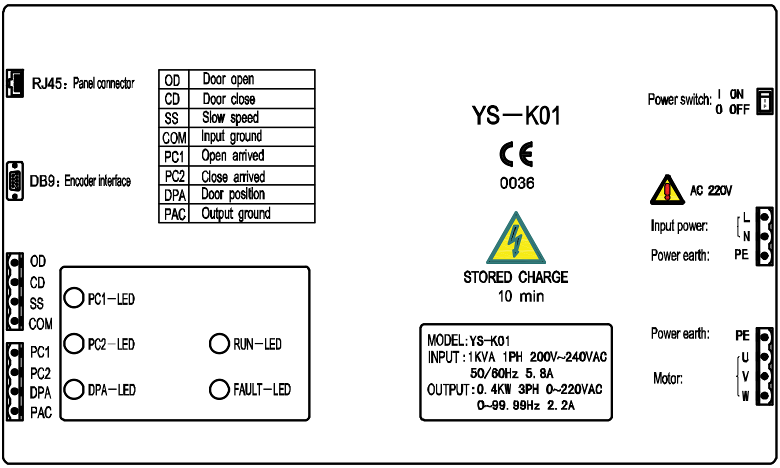


Figure 1-1 Position of terminal

1.1.1 Power Supply Switch

In order to make debugging and maintenance work easily, we have designed a power switch in the controller which will help engineers to operate elevator easily.

Note:

1. When the power supply of controller is cut down, the input 220V terminal still has high voltage. Do not touch or plug the 220V terminal until the external power supply cut down, or there will be danger of shocking.

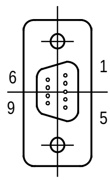
1.1.2 Input Main Circuit Power Terminal Description

Terminal name	Function description
L、N	Single-phase AC 220V power supply input terminals
PE	Protective earth point

1.1.3 Output Main Circuit Power Terminal Description

Terminal name	Function description
PE	Protective earth point
U、V、W	Motor connection terminal

1.1.4 Control Signal Terminal Description

<table>
<tr><th>Symbol</th><th>Terminal name</th><th>Function description</th></tr>
<tr><td>DB9</td><td>Pulse encoder signal terminal</td><td>+24V, the maximum output current is 200mA; and the maximum pulse frequency is 35kHz


<table>
<tr><th>DB9 pin No.</th><th>Signal identity</th></tr>
<tr><td>1</td><td>COM</td></tr>
<tr><td>2</td><td>A</td></tr>
<tr><td>3</td><td>/Z</td></tr>
<tr><td>5</td><td>+24</td></tr>
<tr><td>6</td><td>B</td></tr>
<tr><td>7</td><td>Z</td></tr>
<tr><td>4、8、9</td><td>Reserved</td></tr>
</table></td></tr>
<tr><td>OD</td><td>Open-door command input terminal</td><td rowspan="3">The ON command is valid when terminal is connected with COM
The OFF command is invalid when terminal is disconnected with COM</td></tr>
<tr><td>CD</td><td>Closed-door command input terminal</td></tr>
<tr><td>SS</td><td>Slow speed command (Multi function optional)</td></tr>
<tr><td>COM</td><td>Input signal ground</td><td>Isolated from output ground PAC</td></tr>
<tr><td>PC1</td><td>Door open arrival signal output terminal</td><td rowspan="2">Contact rating: 125VAC/0.5A or 24VDC/1A
Relay output. It can be set as low or high level via parameter</td></tr>
<tr><td>PC2</td><td>Door close arrival signal</td></tr>
</table>

	output terminal	DOA,CME normally open; DCA,CME normally open; DPA,CME normally open
DPA	Door position signal output terminal	
PAC	Output signal ground	Isolated from input ground COM

1.1.5 Panel Interface Description

The controller can be connected to the optional special panel (YS-P01, YS-P02 or YS-P03) through RJ45 interface by general network wires. It can be used for the user parameter settings, copy and operation monitoring state etc. RJ45 interface is described as Figure 1-2.

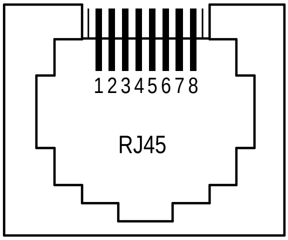


Figure 1-2 Panel interface

1.1.6 State Indicator Description

Symbol	Description
PC1-LED (green)	Set the open-door arrival as normally open contact and the indicator is off. When open door arrives, the indicator will be on Set the open-door arrival as normally closed contact and the indicator is on. When open door arrives, the indicator will be off
PC2-LED (green)	Set the closed-door arrival as normally open contact and the indicator is off. When closed door arrives, the indicator will be on Set the closed-door arrival as normally closed contact and the indicator is on. When closed door arrives, the indicator will be off
DPA-LED (green)	Set the door position signal as normally open contact and the indicator is off. When position arrives, the indicator will be on Set the door position signal as normally closed contact and the indicator is on. When position arrives, the indicator will be off
RUN-LED (green)	When the controller is not in running state at power on (standby), indicator will be on When the controller is in running state at power on, indicator will be flashing
FAULT-LED (red)	When the controller has fault, indicator will be on When the controller has no fault, indicator will be off

1.2 Terminal Wiring

The typical connection of controller is shown as Figure 1-3.

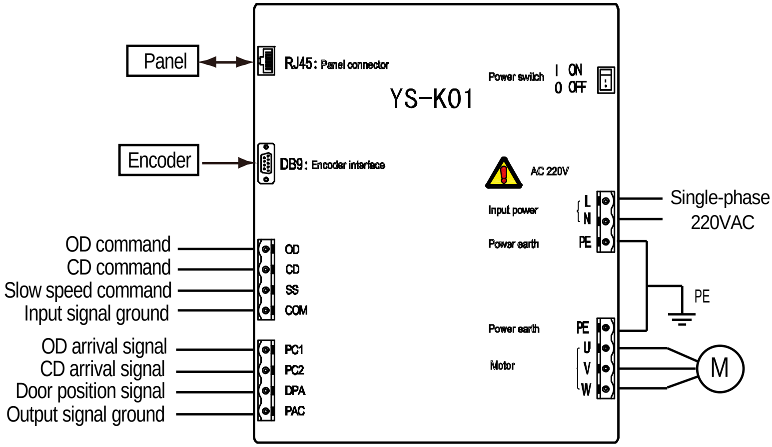


Figure 1-3 Controller connection

Note:

1. The input MCCB rating should use 6A single-phase air breaker.

2. The input power wiring and the output motor wiring should use copper multi-stranded cable whose diameter should not be less than 1mm^2.

3. It suggested that reliably connect the PE terminal to the ground via using copper multi-stranded cable whose diameter should not be less than 2.5mm^2.

4. Control signal input &output wires are suggested to use copper multi-stranded cable (diameter ⩾ 0.5). On serious interference condition you can use twisted-pair or shielded cable to improve the control system capacity.

Chapter 2 Operation Instructions

2.1 YS-P02 Panel Description

The default parameters of the controller can be changed by the optional LED panel (YS-P02). The YS-P02 is shown as Figure 2-1.

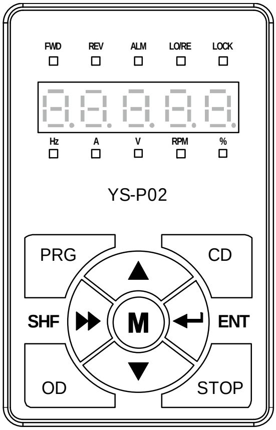


Figure 2-1 YS-P02 panel

There are keys on the YS-P02 panel and their functions, as shown in Table 2-1.

Table 2-1　Description of keys of YS-P02

Key	Name	Function
PRG	Programmable/back	Entry or exit programming key
OD	OD key	OD running command
CD	CD key	CD running command
STOP	Stop/reset key	To stop running; and to manually reset the fault
M	Multi-function key	Reserved
↵	ENT enter/confirm key	Confirm saving the data
▶▶	SHF shift key	Selecting display parameter and shift bit
▲	Increment key	Increase value or parameter
▼	Decrement key	Decrease value or parameter

The YS-P02 panel has 5-digit LED digital tubes. But the left LED digital tube of YS-P02 will not display all the time (except when the initial power on and the panel self-testing).

The YS-P02 panel has 5 state and 5 unit indicators and their meanings are shown as Table 2-2.

Table 2-2 Indicator description of YS-P02

Identification	Name	Description
FWD	OD run state indicator	ON: The present controller is in OD state OFF: The present controller is not in OD state
REV	CD run state indicator	ON: The present controller is in CD state OFF: The present controller is not in CD state
ALM	Warning state indicator	ON: The present controller is faulty OFF: The present controller is not faulty
LO/RE	Remote/local state indicator	ON: The present controller is not in panel control mode OFF: The present controller is in panel control mode
LOCK	Password lock state indicator	ON: The user password of present controller is enabled OFF: The present controller has no password or is at unlock state
Hz	Frequency unit indicator	ON: The present unit is Hz OFF: The present unit is not Hz
A	Current unit indicator	ON: The present unit is A OFF: The present unit is not A
V	Voltage unit indicator	ON: The present unit is V OFF: The present unit is not V
RPM	Rpm unit indicator	ON: The present unit is rpm OFF: The present unit is not rpm
%	Percentage unit indicator	ON: The present unit is % OFF: The present unit is not %

The indicator has two states off and on. In the manual the identification is as follows:

☐ The indicator is off; ■ The indicator is on.

2.1.1 Display State of YS-P02 Panel

The controller parameters can be displayed via using the optional YS-P02 panel. The display state of panel of controller has stopping state, running state, editing state and alarming state.

Display parameter at stopping state

When controller stops running, the panel will display parameter at stopping state, as shown in Figure 2-2. By pressing ▶▶ key to display other stopping parameters: door position pulse, OD/CD arrival signal (OD pulse arrival, OD torque arrival, CD pulse arrival, CD torque arrival), setting frequency, input terminal state and DC bus voltage.

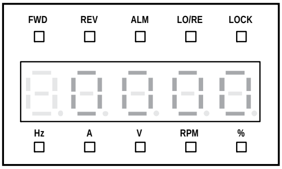


Figure 2-2 YS-P02 display at stopping state

Display parameter at running state

When the controller is at running state, the panel will display the running state parameters which are shown as Figure 2-3. Press ▶▶ key to display other running parameters: door position pulse, OD/CD arrival signal (OD pulse arrival, OD torque arrival, CD pulse arrival, CD torque arrival), setting frequency, output frequency, output voltage, output current, output torque, DC bus voltage and input terminal state.

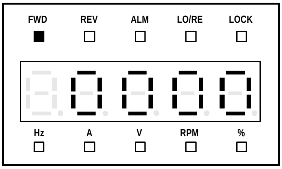


Figure 2-3 YS-P02 display at running state

Remark:

Display the input terminal states in hex and each bit (binary) of this function parameter stands for different physical sources, as following table.

Tens				Units			
Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
-	-	-	-	-	SS	CD	OD

0: The corresponding terminal and the common terminal are disconnected;

1: The corresponding terminal and the common terminal are connected.

Display parameter at editing state

At stop, run or fault alarm state, press PRG to enter function parameter editing state.

The panel uses third-level menu configuration for parameter setting or other operations. In turn: function parameter group setting (first-level menu)→function parameter setting (second-level menu)→parameter setting (third-level menu). Which is shown in Figure 2-4 and the description of the keys is shown in Table 2-3.

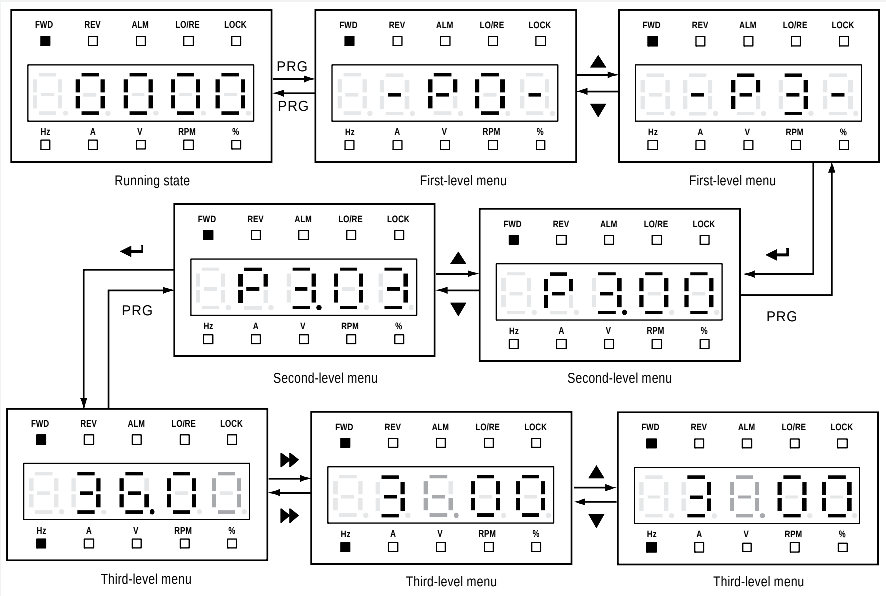


Figure 2-4 YS-P02 parameter editing state

Table 2-3 Switching description of keys of YS-P02

Key	First-level menu	Second-level menu	Third-level menu
PRG	Fault, return to faulty display; Fault cleared, return to run or stop state display.	Return to first-level menu	Do not save the present value and return to second-level menu
↵	Enter to second-level menu	Enter to third-level menu	Save the present value and return to second-level menu
▲	Select function group	Modify the internal No. of function group. Increase by 1 according to the present modified bit	Modify function value. Increase by 1 according to the present modified bit
▼	Select function group	Modify the internal No. of function group. Decrease by 1 according to the present modified bit	Modify function value. Decrease by 1 according to the present modified bit
▶▶	Invalid	Invalid	Switch units, thousands, hundreds, tens

Display faulty state

If the controller is faulty, the panel will display the fault code. After troubleshooting, press STOP key to reset and the panel restores to stop state. Refer to Chapter 4.

Chapter 3 Controller Debugging

Notice:

1. In factory, the controller has been set the parameters. In generally the user don`t need to change parameters.

2. Default control mode is terminal mode (P0.06=1), if you need to use manual mode (press OD/CD to open/close door), please set P0.06=0.

3. Default door arrived **contact** is normally closed (P6.00=1), if you need normally opened, set P6.00=0.

4. The situation that you can`t changing the parameters:

A: Make sure you input the right password, default with no password. Check chapter4 or contact the factory for help if you forget the password. The lock LED on panel YS-P02 can help you to judge whether the controller has password.

B: When the motor is running, you can`t changing some parameters.

a: In terminal mode(P0.06=1), you can pull out the OC/CD terminal, and reset the power.

b: In manual mode(P0.06=0), you can press OD /CD, when the door in the middle place , press stop, then you can change parameters.

3.1 MS debugging steps

You can use the following steps when you change a controller or mistaken parameters.

Step 1: remove the motor belt, set P8.10 according to below table, press enter, let the door do motor learning. It may take 20~30 seconds.

For example: Two panel centre opening 800, set P8.10=31, press enter.

Two panels centre opening	P8.10	Four panels Centre opening	P8.10	Six panels Centre opening	P8.10	One panels Side opening	P8.10	two panels Side opening	P8.10	Three Panels Side opening	P8.10
600	31	1200	31	1800	32	600	31	600	31	900	31
700	31	1300	31	1900	42	700	31	700	31	1000	31
800	31	1400	31	2000	42	800	31	800	31	1100	31
900	31	1500	32	2100	42	900	31	900	31	1200	31
1000	31	1600	32	2200	42	1000	31	1000	31	1300	31
1100	31	1700	32	2300	42	1100	31	1100	31	1400	31
1200	31	1800	32	2400	42	1200	31	1200	31	1500	32
1300	31	1900	42	2500	42	1300	31	1300	31	1600	32
1400	31	2000	42	2600	42	1400	31	1400	31	1700	32
1500	32	2100	42	2700	42	1500	32	1500	32	1800	32
		2200	42	2800	42			1600	32	1900	42
		2300	42	2900	42			1700	32	2000	42
		2400	42	3000	42			1800	32	2100	42
		2500	42	3100	42			1900	42	2200	43
		2600	42	3200	42			2000	42	2300	43
		2700	42	3300	42					2400	43
		2800	42	3400	42					2500	43
		2900	42	3500	42						
		3000	42	3600	43						
		3100	42	3700	43						
		3200	42	3800	43						
		3300	43	3900	43						
		3400	43	4000	43						
		3500	43	4100	43						

Step 2: Set P8.10=99;

Press enter, do door width self-learning. The door will do open-close-open-close (If the door close first, you need to change any two phase of UWV of motor and redo step2), and the door-width will be stored in P2.06. Controller will call parameters according to the door width. Finally it will change to terminal mode P0.06=1, let the system control the door.

Generally you just need to do the above two steps before you running the car door by elevator signal.

3.1.1 OD Curve Optimal

If you are not satisfied with the default OD/CD curve, you can change the relevant parameters according to Figure 3-1 /3-2

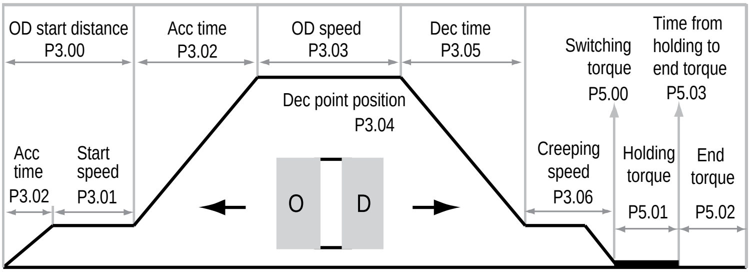


Figure 3-1 OD curve and curve parameters

OD start: From closed state to the OD startup, the door blade needs action. You need use a lower speed (P3.01) to close the door blade in order not to damage it. After the startup state ends, it will accelerate to high-speed for OD.

OD high-speed run: After OD startup ends, OD will accelerate from start speed (P3.01) to OD speed (P3.03) The run speed will not decelerate to creeping speed (P3.06) until it runs to Dec point (P3.04).

OD creep: To prevent door and other mechanical equipment from collision damage caused by high-speed, before OD position arrives, door should slow down to creeping speed (P3.06).

OD holding: After door creeps through the OD limit point (P3.07), it will output the OD arrival signal. When door torque＞OD switching torque (P5.00), it will switch to OD torque holding state.

After some time the door is in the fully open state, then switch to a small torque holding state, which can save energy and prevent motor from heating to ensure that the door has certain open tension.

3.1.2 CD Curve Optimal

A full itinerary CD curve has five stages: CD start, CD high-speed run, CD creep, door blade action and CD holding. According to the Figure 3-2 we can set reasonable parameters to achieve the desired CD effect.

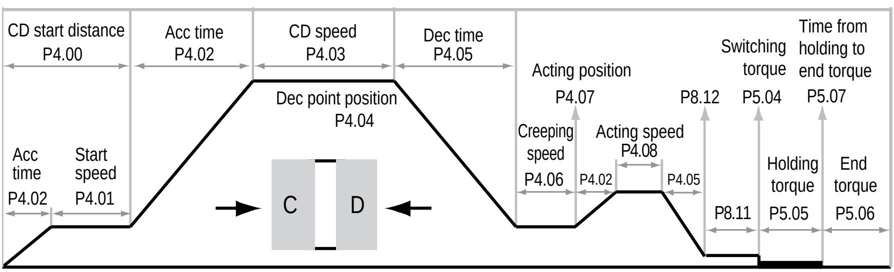


Figure 3-2 Relationship of CD curve and curve parameters

CD start: The CD starts to run at start speed (P4.01).

CD high-speed run: After CD startup ends, CD will accelerate from CD start speed (P4.01) to CD speed (P4.03). The run speed will not decelerate to creeping speed (P4.06) until it run to Dec point (P4.04).

CD creep: Before CD position arrives, to prevent door board and door blade from collision damage caused by high speed, the door should slow down to creeping speed to run.

Door blade action: From CD process to two door boards close state, a short time relative higher than creeping speed (P4.08) is needed to achieve door blade action safe and reliable as soon as possible, while it can improve the CD efficiency.

At the door blade action end, to prevent door crash because of high-speed, you can set a relative low speed via P8.11 and this speed range in pulses of P8.12 is valid.

CD holding: After door creeps through the CD limit point (P4.09), it will output CD arrival signal. When CD torque＞switching torque (P5.04), it will switch to CD torque holding state.

After CD holds for some time, the door has been in the fully closed state. Then switch to a small torque holding state, which can prevent motor from heating and save energy, therefore ensure that the door has certain closed tension.

3.2 Checking encoder

Once you can`t make the door running well, check the encoder first. Pull out OD/CD/SS/COM terminal. Reset the power of controller, first screen of YS-P02 is encoder pulse, move the door, the encoder pulse should increase in open direction and reduce in close direction about 200/circle.

Notice: the encoder pulse of first screen cannot be bigger than the parameter in P2.06, so you can set a big number in P2.06 and try again.

3.3 Closing force Adjustment

You can reduce P8.13 (reacting time of reopen) and P5.08 (reopen torque) to adjust the closing force.

Chapter 4 Trouble shooting

1. When fault alarm occurs, please take proper action according to the Table 4-1.

Table 4-1 Fault alarm description and counter-measures

Fault code	Fault name	Possible reasons of fault	Counter-measures
-Lu-	DC bus undervoltage	• At the begining of power on and at the end of power off • Input voltage is too low • Improper wiring leads to undervoltage of hardware	• It is normal state of power on and power off • Please check input power voltage • Please check wiring and wire properly
E001	DC bus overvoltage	• Input voltage is too high • Deceleartion time is too short • Improper wiring leads to overvoltage of hardware	• Please check power input • Please set a proper value for Dec time • Please check wiring and wire properly
E002	Controller output instantaneous overcurrent (hardware)	• Improper connection between controller and motor • Improper motor parameters • The encoder signal error • Improper wiring leads to overcurrent of hardware • Acceleration/deceleration time is too short	• Connect the controller and motor properly • Please set correct motor parameters • To ensure correct encoder signal, check the encoder connection • Please check system wiring and wire properly • Please set proper Acc time and Dec time
E003	Controller output overcurrent (software)		
E004	Loss of Z phase signal of permanent magnet synchronous motor encoder	• Without detecting the Z signal	• Detect the encoder signal
E005	Machine type mismatch	• YS-P01 version is too old (below V1.02)	• Press stop,then you can operate it, once you finished,take off YS-P01 • Alarm fault, the panel should be reset one time • Use the new panel YS-P02 or YS-P03
E006	Motor parameter auto-tuning fault	• Motor rated parameters improperly set • Motor incorrect connection • Encoder Z signal loss	• Set proper motor parameters • Please check the motor's connection • Please check the encoder
E007	Controller overloaded	• The load is too big • There are obstacles in door machine	• Regulate the mechanical • Check machinery, to exclude obstacles

Fault code	Fault name	Possible reasons of fault	Counter-measures
E008	Heatsink overheated	• The temperature detection circuit is abnormal • The outer vent is jammed	• Please contact the supplier for repairing • Check the outer vent
E009	Motor heat protection	• Motor current is too big for a long time	
E010	Reserved		
E011	Panel EEPROM faulty	• Memory circuit fault of panel EEPROM	• Replace the panel • Please contact the supplier for repairing
E012	Reserved		
E013	Communication fault between the panel and the controller	• RJ45 connection error • RJ45 connection is broken off or loose	• Check the connection, restart to plug the panel • Check the connection, restart to plug the panel
E014	Controller EEPROM faulty	• Memory circuit fault of controller EEPROM	• Please contact the supplier for repairing
E015/E016	Reserved		
E017	Module faulty	• Short circuit between phases output or the ground • Short circuit for the ground • Output current is too high	• Please check the connection and connect the wire properly • Please check the connection and connect the wire properly • Please check the connection and mechanism
E018	Current detect faulty	• Current detection circuit is damaged	• Please contact the supplier for repairing
E019~E022	Reserved		
E023	Encoder error	• The damaged encoder • The wrong connection of encoder • Wrong setting of encoder pulse number per revolution	• Check the encoder, replace it • Check the encoder connection and wire correctly • Please set correct value of pulse number per revolution (P2.00)
E024/E025	Reserved		
E026	Door width self-learning failure	• The faulty encoder • The wrong connection of encoder • Improper setting self-learning parameters of speed or torque	• Check the encoder, replace it • Check the encoder connection and wire correctly • Please set correct value of self-learning speed (P2.04) and self-learning torque (P5.10)

2. If abnormal situation occur, please take proper action according to Table 4-2.

Table 4-2 Abnormal situation and counter-measures

Number	Abnormal situation	Counter-measures
1	Forget the password	• Clear the password, set P0.00=2801,press ENT,go back to P000, when 0000 the last 0 is shining, press ENT.
2	Can`t see some parameters	• In put the advanced password P0.08=9888 Please don`t change the advanced parameters ,just use the default parameters.
3	The door can`t run by the siginal from lift.	• Pull out wires in OD/CD/SS/COM terminal, tighten the screws,use a short wire to connect OD/COM,CD/COM. If it can close and open the door, check the outside system please. • Reset paramenters, and try again
4	The door always reopen in nearly closed position	• Move the control plug(OD/CD/SS/COM),pull the door out of the holdding area, use your hand pull the belt to open and close direction in this reopen position. If it`s stuck on mechanical, check the mechanic prat or call factory for help, if not ,you can increase P5.08 like 10~20% percent
5	In terminal mode, after the door open arrived, it did not close, or after close arrived, it did not open	• Check contact type of OPEN/CLOSE arrived siginal, default P6.00=1, normally closed, if you need normally open, set P6.00=0

Chapter 5 Parameters table

Attributes are changed:

"*": It denotes that the value of this parameter is the actual value which cannot be modified.

"×": It denotes that the setting parameter cannot be modified in run state.

"○": It denotes that the setting parameter can be modified in run state.

Code	Name	Range	MS	Attributes Setting	
Group P0 Basic Parameters					
P0.00	User password	0000—9999	0000	○	
P0.01	Parameter batch disposal selection	00: No function. It need manually set the parameter 01: Upload parameters 02: Download parameters 03: Reset default parameter, synchronous motor MS parameters 04: Clear fault information.	00	×	
P0.02	Parameter upload number	1—12	1	×	
P0.03	Parameter download number	1—12	1	×	
P0.04	Parameter batch disposal comsirm	0: No function 1: Press ↵ key to comsirm the batch disposal. This parameter (P0.04) will restore to zero when the parameter batch disposal is finished	0	×	
P0.05	Mode selection	0: Factory mode (door machine specific function is invalid) 1: Door control mode	1	×	
P0.06	Control command selection	0: Panel control 1: Terminal control (auto) 2: Auto-demo run	1	×	
P0.07	Maximum output frequency	1.00—99.99Hz	50.00Hz	×	
P0.08	Advanced password	After input the advanced password, you can check the advanced parameters	9888	○	
P0.09	Control mode	0: Asynchoronous motor open-loop distance control 1: Asynchoronous motor closed-loop distance control 2: Synchoronous motor closed-loop distance control	1	×	
Group P1 Motor Parameters					
P1.00	Motor rated power	1—750W, default 250W	250W	×	
P1.01	Motor rated voltage	1—300V, default 220V	220V	×	
P1.02	Motor rated current	0.10—10.00A, according to motor nameplate's rated current	0.55/0.90	×	

Code	Name	Range	MS	Attributes	Setting
P1.03	Motor rated frequency	1.00—P0.07, according to motor nameplate's rated frequency	50.00Hz	×	
P1.04	Motor rated Rpm	1—6000rpm, according to motor nameplate's rated rpm	900rpm	×	
P1.05	Deceleration ratio	1.00—9.99	4.67/5.60	×	
P1.06	Motor parameter auto-tuning	0: Disabled 1: Motor auto-tuning	0	×	
P1.07	No load current	0.01—10.00A	0.30/0.60	×	
P1.08	Reserved				
P1.09	Magnetic pole position angle	0.0—359.9°	0.0°	×	
Group P2 Encoder and Door-Width Parameters					
P2.00	Encoder pulse number per revolution	0—9999	200	×	
P2.01	Encoder direction setting	0: The same direction as the encoder actual connection 1: The reverse direction to the encoder actual connection	1	×	
P2.02	Operating speed of slow speed	0.01—15.00Hz	7.00Hz	○	
P2.03	Operating speed at first time power on	0.01—15.00Hz	7.00Hz	○	
P2.04	Door width self-learning speed	0.01—15.00Hz	7.00Hz	○	
P2.05	Door width self-learning enable	0: Disabled 1: Door width self-learning is enabled	0	×	
P2.06	Low digits of pulse count	0—9999 (pulse number)	780	×	
P2.07	High digits of pulse count	0—9999 (pulse number)	0	×	
Group P3 Door-Open Curve Parameters					
P3.00	OD start distance	0—9999 (pulse number)	90	○	
P3.01	OD start speed	0.00—15.00Hz	4.00Hz	○	
P3.02	OD Acc time	0.1—99.9s, recommended value is 0.8—2.0s	1.0s	○	
P3.03	OD speed	0.00—max frequency (P0.07), recommended value is 30—45Hz	32.00Hz	○	
P3.04	OD Dec point position	50.0—90.0% (door width), recommended value is 65—85%	70%	○	
P3.05	OD Dec time	0.1—99.9s, recommended value is 0.8—2.0s, it will properly increase with the door width increasing	1.3s	○	
P3.06	Creeping speed at OD ending	0.00—15.00Hz	4.0Hz	○	
P3.07	OD limit point	80.0—99.9% door width	95.0%	○	
P3.08	Re-open curve high-speed	(from OD speed to start Dec) 10.0—90.0%	90.0%	○	

Code	Name	Range	MS	Attributes Setting	
	area	(door width)			
Group P4 Door-Close Curve Parameters					
P4.00	CD start distance	0—9999 (pulse number)	0	○	
P4.01	CD start speed	0.00—15.00Hz	4.00Hz	○	
P4.02	CD Acc time	0.1—99.9s, recommended value is 0.8—2.0s, it will properly increase with the door width increasing	2.0s	○	
P4.03	CD speed	0.00—max frequency (P0.07), recommended value is 26—38Hz	24.00Hz	○	
P4.04	CD Dec point position	50.0—90.0% (door width), recommended value is 65—85%	72.0%	○	
P4.05	CD Dec time	0.1—99.9s, recommended value is 0.8—2.0s, it will properly increase with the door width increasing	1.3s	○	
P4.06	Creeping speed at CD ending	0.00—15.00Hz	3.50Hz	○	
P4.07	Door blade acting position at CD ending	0—210 (pulse number)	80	○	
P4.08	Door blade acting speed at CD ending	0.00—15.00Hz	5.50Hz	○	
P4.09	CD limit point	0—200 (pulse number)	60	○	
Group P5 Torque Parameters					
P5.00	OD switching torque	20.0—150.0% (motor rated torque)	60.0%	○	
P5.01	OD holding torque	30.0—150.0% (motor rated torque)	50.0%	○	
P5.02	OD end torque	0.0—100.0% (motor rated torque)	45.0%	○	
P5.03	Switching time from OD holding to OD end torque	0.1—999.9s (the internal has slope)	10.0s	○	
P5.04	CD switching torque	20.0—150.0% (motor rated torque)	50.0%	○	
P5.05	CD holding torque	30.0—150.0% (motor rated torque)	45.0%	○	
P5.06	CD end torque	0.0—100.0% (motor rated torque)	40.0%	○	
P5.07	Time from CD holding to CD end torque	0.1—999.9s (the internal has slope)	3.0s	○	
P5.08	Passenger protection torque	0.0—150.0% The smaller the value is, the more sensitivity it is. If set P5.08 as 0, this function is disabled. At low speed, the setting value increase by 120.0% (motor rated torque)	50.0%	○	
P5.09	Passenger protection invalidation area	0.0—50.0% (door width)	1.0%	○	
P5.10	Torque setting under slow speed	30.0%—150.0% (motor rated torque) Remark: The judgment of door width self-learning and torque switching value at first time powe-on	100.0%	○	
P5.11	OD resistance torque	0.0—150.0%	0%	○	

Code	Name	Range	MS	Attributes Setting	
		The smaller the value is, the more sensitivity it is. If set P5.11 as 0, this function is disabled. At low speed, the setting value increase by 120.0% (motor rated torque)			
P5.12	OD resistance time	0.1—9.9s, this function code defines after the setting time of P5.12, it will respond the OD command at OD resistance	3.0s	○	
Group P6 Advanced Parameters					
P6.00	PC1, PC2, DPA output selection	0: The signal is open which represents the OD/CD arrival and door position output (equal to the relay normally open contact) 1: The signal is closed which represents the OD/CD arrival and door position output (equal to the relay normally closed contact).	1	×	
P6.01	DPA function selection	0: Door position output 1: The re-open signal is output at the CD resistance	0	×	
P6.02	Door position output setting	0.1—99.9%	90%	○	
P6.03	Dec time of open & close resistance	0.1—2.0s	0.5s	○	
P6.04	Open & close holding time on demo mode	0.1—360.0s	3.0s	○	
P6.05	Mode selection on open & close invalidation	0: Holding the torque at the OD&CD arrival range, but zero-speed running at other position 1: Stop running 2: Only holding the torque at the OD&CD arrival range	2	○	
P6.06	Torque setting on open&close invalidation	0—150.0%. When set P6.05 as 0, the special torque will be output to prevent the door from moving	100.0%	○	
Group P6 Vector Control Parameters					
P7.00	High speed ASR KP	10—3000	1000	○	
P7.01	High speed ASR KI	0 (invalid), 1—1000	50	○	
P7.02	Low speed ASR KP	10—3000	1000	○	
P7.03	Low speed ASR KI	0 (invalid), 1—1000	50	○	
P7.04	ASR switching frequency	0.00—P0.07	8.00Hz	○	
P7.05	Current loop KP	10—9999	100	○	
P7.06	Current loop KI	0—9999	800	○	
P7.07	Torque limit	0.0—200.0% (motor rated current)	120.0%	×	
P7.08	Speed filter time constant	0—7	2	○	
Group P8 Advanced Parameters					
P8.00	Stator resistance	0.00—99.99Ω	Motor decide	×	
P8.01	Rotor resistance	0.00—99.99Ω		×	

Code	Name	Range	MS	Attributes	Setting
P8.02	Stator inductance	0—9999mH		×	
P8.03	Rotor inductance	0—9999mH		×	
P8.04	Mutual inductance	0—9999mH		×	
P8.05	Slip compensation gain	50.0—200.0%	100.0%	×	
P8.06	Curve selection	0: Line 1: S curve	0	○	
P8.07	LCD contrast	0—10	0	○	
P8.08	LCD screen color only for YS-P03	0: White background 1: Black background	0	○	
P8.09	LCD language for YS-P03	0: Chinese 1: English	0	○	
P8.10	Reset Eshine special parameters	00—99	00	×	
P8.11	CD door blade closed speed	0.00—2.00Hz	0.80Hz	×	
P8.12	CD door blade closed speed position	0—80	30	×	
P8.13	Reacting time for reopen the door	0~400ms	60~120		
Group P9 Fault Record Parameters					
P9.00	NO.3 fault type (last fault)	1: Controller overvoltage (E001) 2: Controller hardware overcurrent (E002) 3: Controller software overcurrent (E003) 4: Loss of Z phase signal of permanent magnet synchronous motor encoder (E004) 5: Machine type mismatch (E005) 6: Motor parameter auto-tuning fault (E006) 7: Controller overloaded (E007) 8: Heatsink overheated (E008) 9: Motor heat protection(E009) 10: Reserved 11: Panel EEPROM faulty (E011) 12: Reserved 13: Communication fault between the panel and the controller (E013) 14: Controller EEPROM faulty (E014) 15,16: Reserved 17: Module faulty (E017) 18: Current detect faulty (E018) 19—22: Reserved 23: Encoder error (E023) 24,25: Reserved 26: Door width self-learning failure (E026) -Lu-: Undervoltage	*	*	
P9.01	NO.2 fault type		*	*	
P9.02	NO.1 fault type		*	*	
P9.03	DC bus voltage for last fault	0—999V	*	*	

Code	Name	Range	MS	Attributes Setting	
P9.04	Output current for last fault	0.00—99.99A	*	*	
P9.05	Running frequency for last fault	0.00—99.99Hz	*	*	
P9.06	Door position for last fault	0—65535	*	*	
P9.07	Open-close cycle lower value	0—9999	*	*	
P9.08	Open-close cycle upper value	0—9999	*	*	
P9.09	Running hour	0—23	*	*	
P9.10	Running day	0—9999	*	*	
P9.11	Heatsink temperature	0.0—150.0℃	*	*	
P9.12	Controller software version	1.00—99.99	*	*	
P9.13	Panel software version	1.00—99.99	*	*	

Chapter 6 Explosion proof door inverter manual

1. In factory, the controller has been set the parameters. In generally the user don`t need to change parameters. You can use the following steps when you change a controller or mistaken parameters.

Step 1: Remove the motor belt, set P8.10=03, press enter

Set P1.02= (Motor current-according to motor nameplate)

If the door width>1800, set P1.05=5.60

Set P0.06=0, P1.06=1, press OD, do motor learning, it may take 20~30 seconds.

Step 2: Put the motor belt, set P8.10=99, press enter to do door width learning. Refer to chapter 3.1 - step2.